LAB 6

Experiment 6: Implementation of Various Join Operations

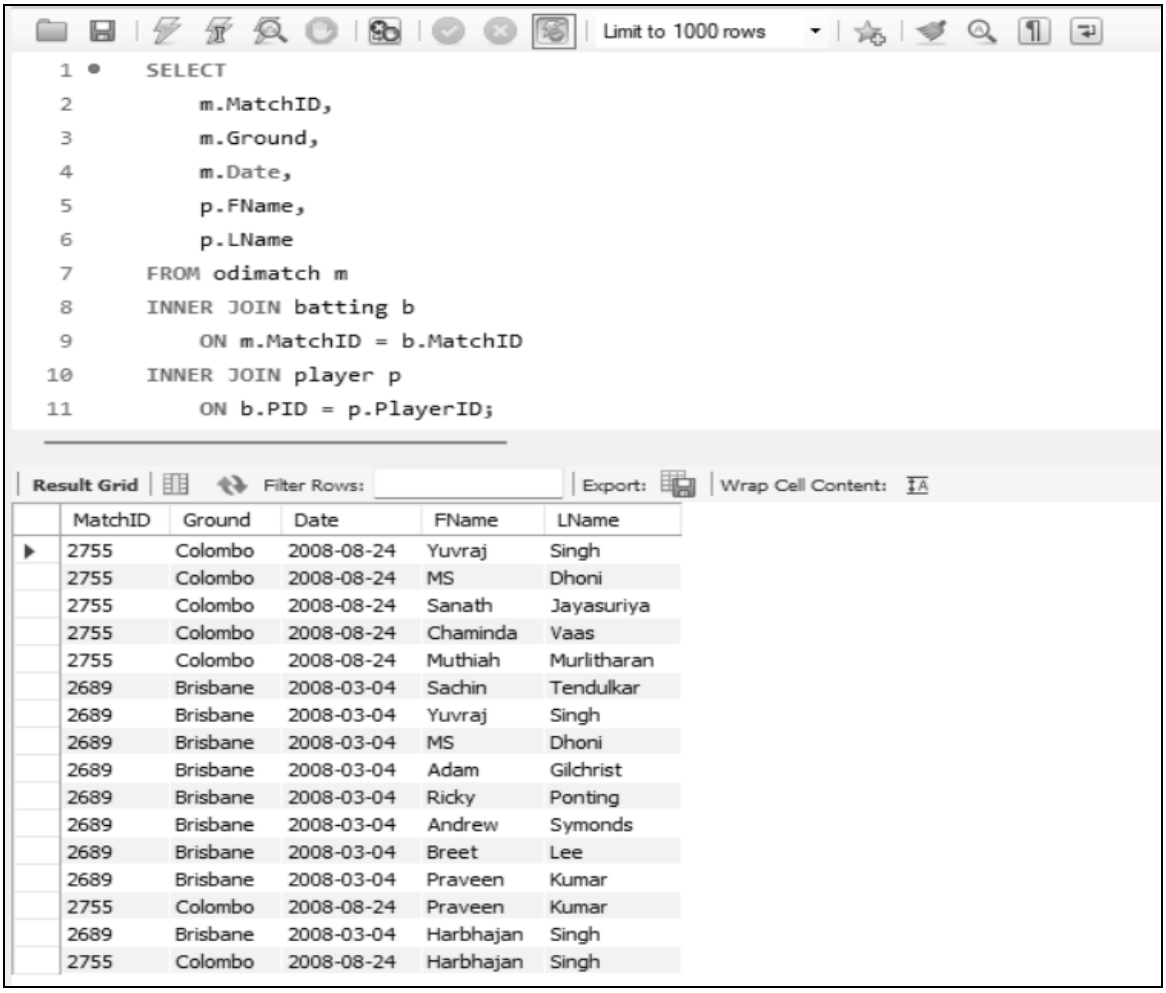
Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

1. Inner Join:

Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

```
USE rr50;

SELECT
m.MatchID,
m.Ground,
m.Date,
p.FName,
p.LName
FROM odimatch m
INNER JOIN batting b
ON m.MatchID = b.MatchID
INNER JOIN player p
ON b.PID = p.PlayerID;
```



```
SELECT
    m.MatchID,
    m.Ground,
    m.Date,
    p.FName,
    p.LName
FROM odimatch m
INNER JOIN batting b
    ON m.MatchID = b.MatchID
INNER JOIN player p
    ON b.PID = p.PlayerID;
```

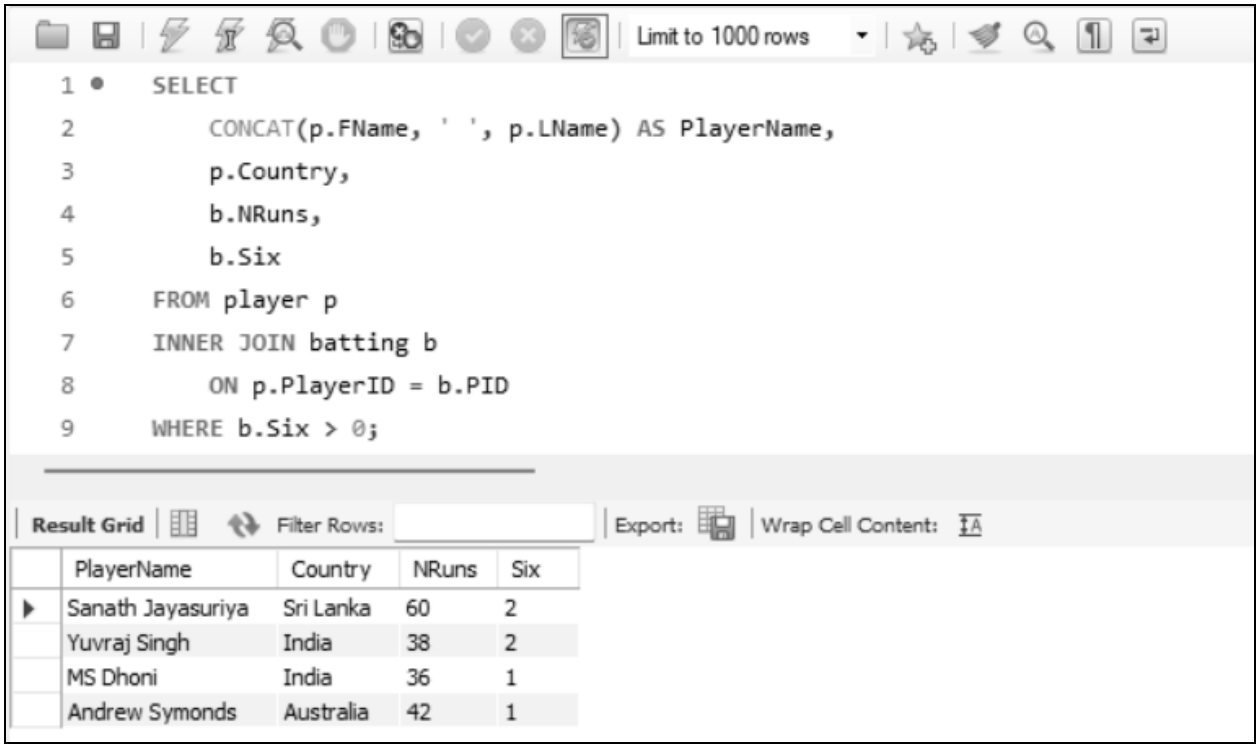
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MatchID	Ground	Date	FName	LName
2755	Colombo	2008-08-24	Yuvraj	Singh
2755	Colombo	2008-08-24	MS	Dhoni
2755	Colombo	2008-08-24	Sanath	Jayasuriya
2755	Colombo	2008-08-24	Chaminda	Vaas
2755	Colombo	2008-08-24	Muthiah	Murlitharan
2689	Brisbane	2008-03-04	Sachin	Tendulkar
2689	Brisbane	2008-03-04	Yuvraj	Singh
2689	Brisbane	2008-03-04	MS	Dhoni
2689	Brisbane	2008-03-04	Adam	Gilchrist
2689	Brisbane	2008-03-04	Ricky	Ponting
2689	Brisbane	2008-03-04	Andrew	Symonds
2689	Brisbane	2008-03-04	Breet	Lee
2689	Brisbane	2008-03-04	Praveen	Kumar
2755	Colombo	2008-08-24	Praveen	Kumar
2689	Brisbane	2008-03-04	Harbhajan	Singh
2755	Colombo	2008-08-24	Harbhajan	Singh

2. Inner Join with Filter:

Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

```
SELECT
CONCAT(p.FName, ' ', p.LName) AS PlayerName,
p.Country,
b.NRuns,
b.Six
FROM player p
INNER JOIN batting b
ON p.PlayerID = b.PID
WHERE b.Six > 0;
```



```
SELECT
    CONCAT(p.FName, ' ', p.LName) AS PlayerName,
    p.Country,
    b.NRuns,
    b.Six
FROM player p
INNER JOIN batting b
    ON p.PlayerID = b.PID
WHERE b.Six > 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerName	Country	NRuns	Six
Sanath Jayasuriya	Sri Lanka	60	2
Yuvraj Singh	India	38	2
MS Dhoni	India	36	1
Andrew Symonds	Australia	42	1

3. Left Outer Join:

Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

```
SELECT
p.PlayerID,
p.FName,
p.LName,
b.MatchID,
b.NRuns
FROM player p
LEFT OUTER JOIN batting b
ON p.PlayerID = b.PID;
```

```
1  SELECT
2      p.PlayerID,
3      p.FName,
4      p.LName,
5      b.MatchID,
6      b.NRuns
7  FROM player p
8  LEFT OUTER JOIN batting b
9      ON p.PlayerID = b.PID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	MatchID	NRuns
21001	Virender	Sehwag	NULL	NULL
23001	Yuvraj	Singh	2755	0
23001	Yuvraj	Singh	2689	38
24001	Andrew	Symonds	2689	42
25001	MS	Dhoni	2755	71
25001	MS	Dhoni	2689	36
27001	Praveen	Kumar	2689	7
27001	Praveen	Kumar	2755	2
27002	Ishant	Sharma	NULL	NULL
89001	Sachin	Tendulkar	2689	91
90001	Brain	Lara	NULL	NULL
90002	Anil	Kumble	NULL	NULL
91001	Sanath	Jayasuriya	2755	60
92001	Shane	Warne	NULL	NULL
92002	Muthiah	Murlitharan	2755	1
92003	Inzamam	UI-Huq	NULL	NULL
93002	Heath	Streak	NULL	NULL
93003	Gary	Kirsten	NULL	NULL
94002	Chaminda	Vaas	2755	17
94004	Stephen	Fleming	NULL	NULL
95001	Ricky	Ponting	2689	1
95002	Shaun	Pollock	NULL	NULL
95003	Jacques	Kallis	NULL	NULL
96001	Rahul	Dravid	NULL	NULL
96002	Herschelle	Gibbs	NULL	NULL
96003	Saurav	Ganguly	NULL	NULL
97004	Daniel	Vettori	NULL	NULL
98001	Shahid	Afridi	NULL	NULL
98002	Harbha...	Singh	2689	3
98002	Harbha...	Singh	2755	2
99001	Breet	Lee	2689	7
99002	Adam	Gilchrist	2689	2
99003	Michale	Vaughan	NULL	NULL
300311	Michale	Clark	NULL	NULL
406317	Sam	Sam	NULL	NULL

4. Right Outer Join:

Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

SELECT

bw.MatchID,

bw.NOvers,

bw.Nwickets,

p.FName,

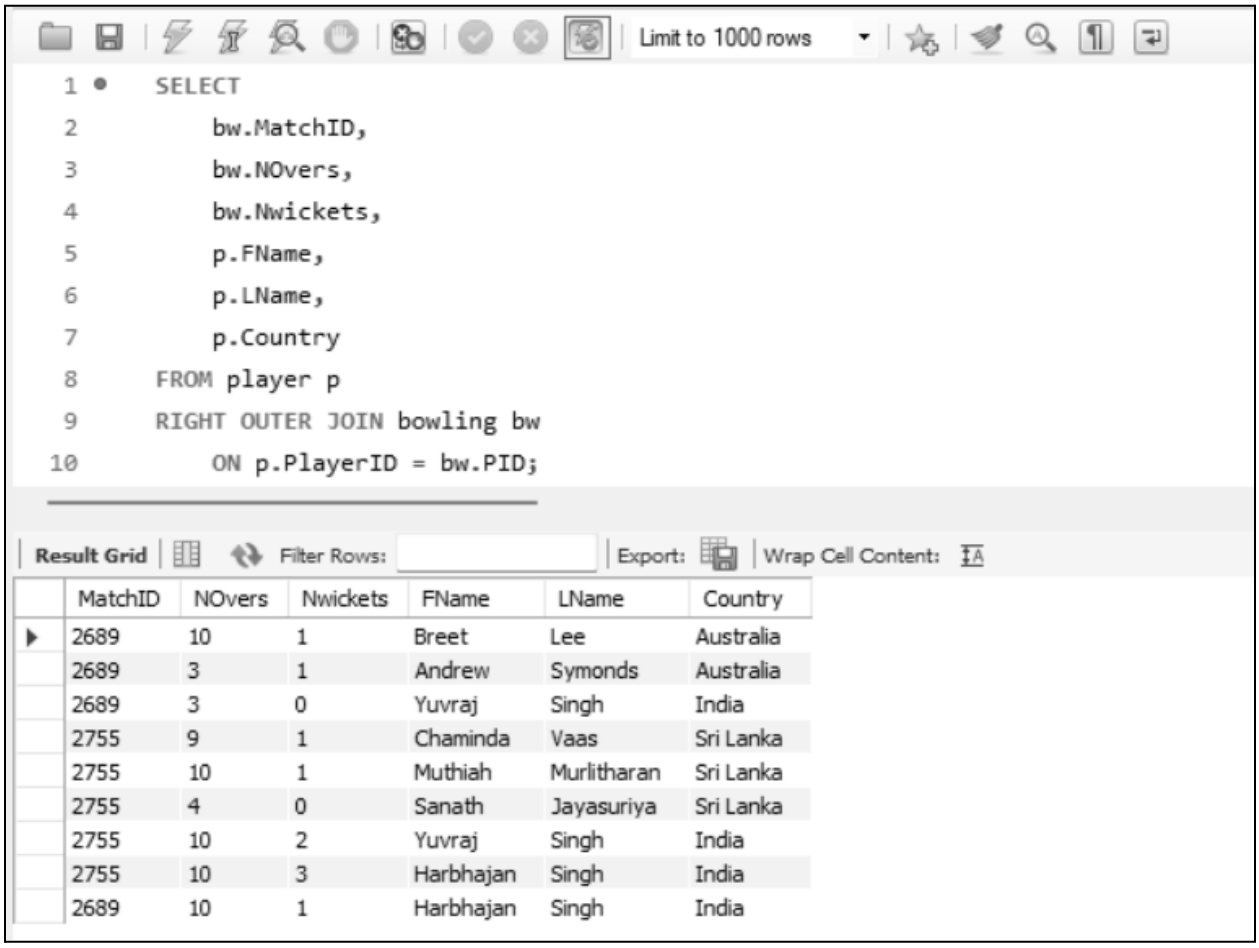
p.LName,

p.Country

FROM player p

RIGHT OUTER JOIN bowling bw

ON p.PlayerID = bw.PID;



MatchID	NOvers	Nwickets	FName	LName	Country
2689	10	1	Breet	Lee	Australia
2689	3	1	Andrew	Symonds	Australia
2689	3	0	Yuvraj	Singh	India
2755	9	1	Chaminda	Vaas	Sri Lanka
2755	10	1	Muthiah	Murlitharan	Sri Lanka
2755	4	0	Sanath	Jayasuriya	Sri Lanka
2755	10	2	Yuvraj	Singh	India
2755	10	3	Harbhajan	Singh	India
2689	10	1	Harbhajan	Singh	India

5. Full Outer Join / Union:

Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

```
SELECT
COALESCE(b.MatchID, bw.MatchID) AS MatchID,
COALESCE(b.PID, bw.PID) AS PID,
b.NRuns,
bw.NOvers,
bw.Nwickets
FROM batting b
LEFT JOIN bowling bw
ON b.MatchID = bw.MatchID
AND b.PID = bw.PID

UNION

SELECT
COALESCE(b.MatchID, bw.MatchID) AS MatchID,
COALESCE(b.PID, bw.PID) AS PID,
b.NRuns,
bw.NOvers,
bw.Nwickets
FROM batting b
RIGHT JOIN bowling bw
ON b.MatchID = bw.MatchID
AND b.PID = bw.PID;
```

```
SELECT
    COALESCE(b.MatchID, bw.MatchID) AS MatchID,
    COALESCE(b.PID, bw.PID) AS PID,
    b.NRuns,
    bw.NOvers,
    bw.Nwickets
FROM batting b
LEFT JOIN bowling bw
    ON b.MatchID = bw.MatchID
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MatchID	PID	NRuns	NOvers	Nwickets
2755	23001	0	10	2
2755	25001	71	NULL	NULL
2755	91001	60	4	0
2755	94002	17	9	1
2755	92002	1	10	1
2689	89001	91	NULL	NULL
2689	23001	38	3	0
2689	25001	36	NULL	NULL
2689	99002	2	NULL	NULL
2689	95001	1	NULL	NULL
2689	24001	42	3	1
2689	99001	7	10	1
2689	27001	7	NULL	NULL
2755	27001	2	NULL	NULL
2689	98002	3	10	1
2755	98002	2	10	3

6. 3-Table Inner Join:

List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

SELECT

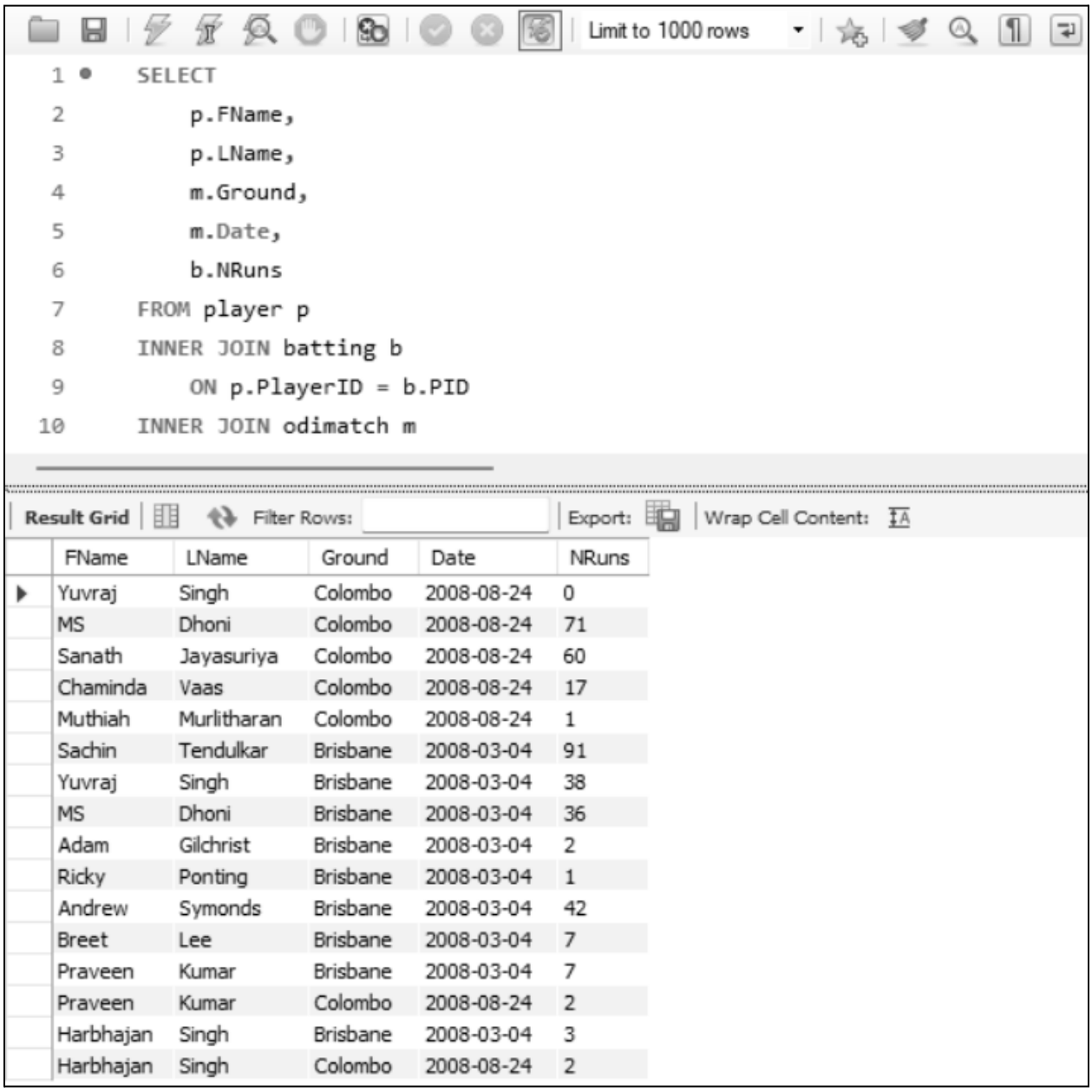
p.FName,

p.LName,

m.Ground,

m.Date,

```
b.NRuns
FROM player p
INNER JOIN batting b
ON p.PlayerID = b.PID
INNER JOIN odimatch m
ON b.MatchID = m.MatchID
WHERE m.Team1 = 'India'
OR m.Team2 = 'India';
```



```
SELECT
    p.FName,
    p.LName,
    m.Ground,
    m.Date,
    b.NRuns
FROM player p
INNER JOIN batting b
    ON p.PlayerID = b.PID
INNER JOIN odimatch m
```

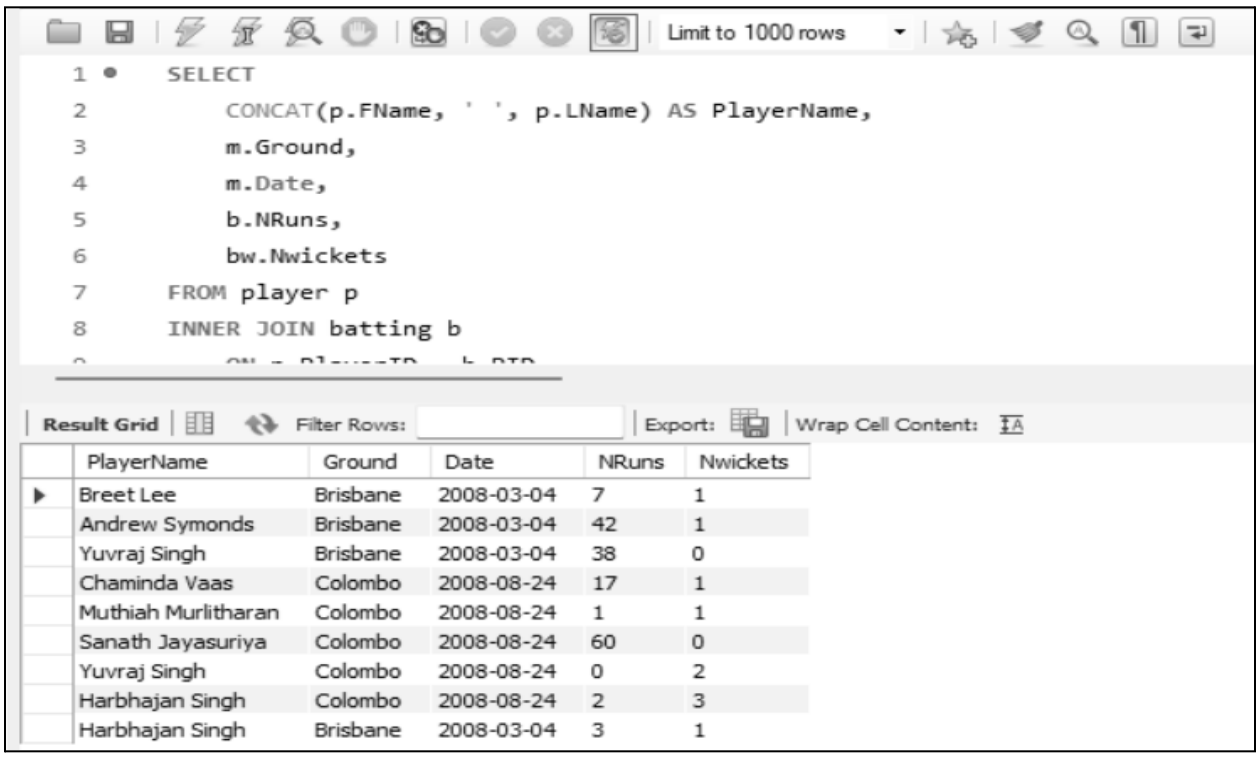
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

7. 4-Table Multi-Join:

Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

```
SELECT
CONCAT(p.FName, ' ', p.LName) AS PlayerName,
m.Ground,
m.Date,
b.NRuns,
bw.Nwickets
FROM player p
INNER JOIN batting b
ON p.PlayerID = b.PID
INNER JOIN bowling bw
ON b.PID = bw.PID
AND b.MatchID = bw.MatchID
INNER JOIN odimatch m
ON b.MatchID = m.MatchID;
```

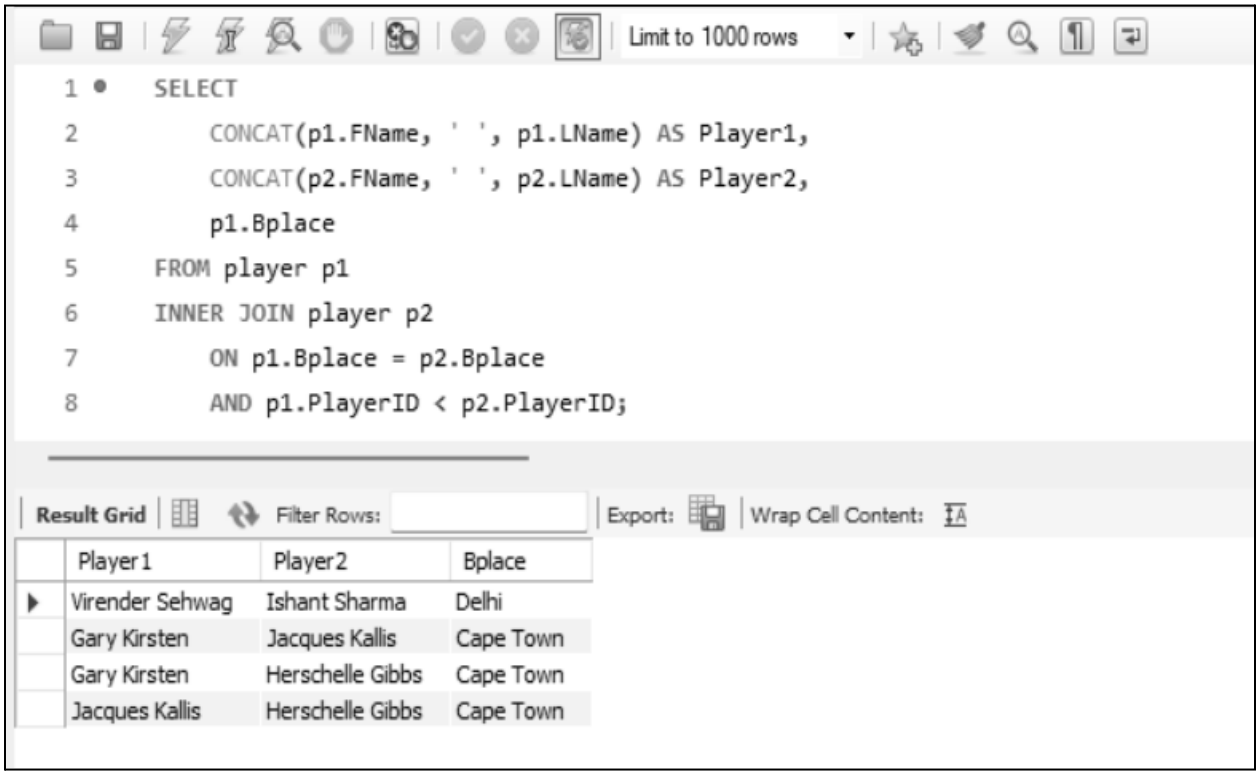


PlayerName	Ground	Date	NRuns	Nwickets
Breet Lee	Brisbane	2008-03-04	7	1
Andrew Symonds	Brisbane	2008-03-04	42	1
Yuvraj Singh	Brisbane	2008-03-04	38	0
Chaminda Vaas	Colombo	2008-08-24	17	1
Muthiah Murlitharan	Colombo	2008-08-24	1	1
Sanath Jayasuriya	Colombo	2008-08-24	60	0
Yuvraj Singh	Colombo	2008-08-24	0	2
Harbhajan Singh	Colombo	2008-08-24	2	3
Harbhajan Singh	Brisbane	2008-03-04	3	1

8. Self Join - Attribute Match:

Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

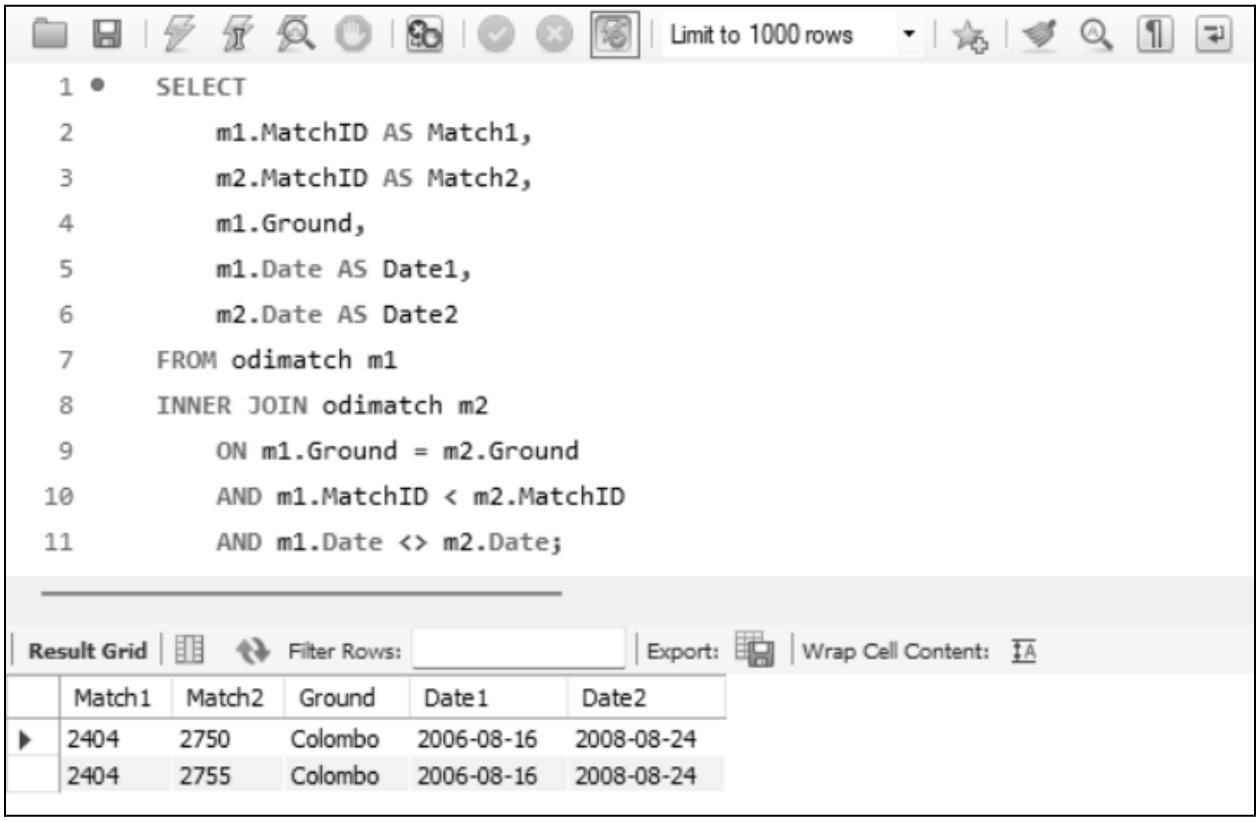
```
SELECT
CONCAT(p1.FName, ' ', p1.LName) AS Player1,
CONCAT(p2.FName, ' ', p2.LName) AS Player2,
p1.Bplace
FROM player p1
INNER JOIN player p2
ON p1.Bplace = p2.Bplace
AND p1.PlayerID < p2.PlayerID;
```



9. Self Join - Match Analysis:

Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

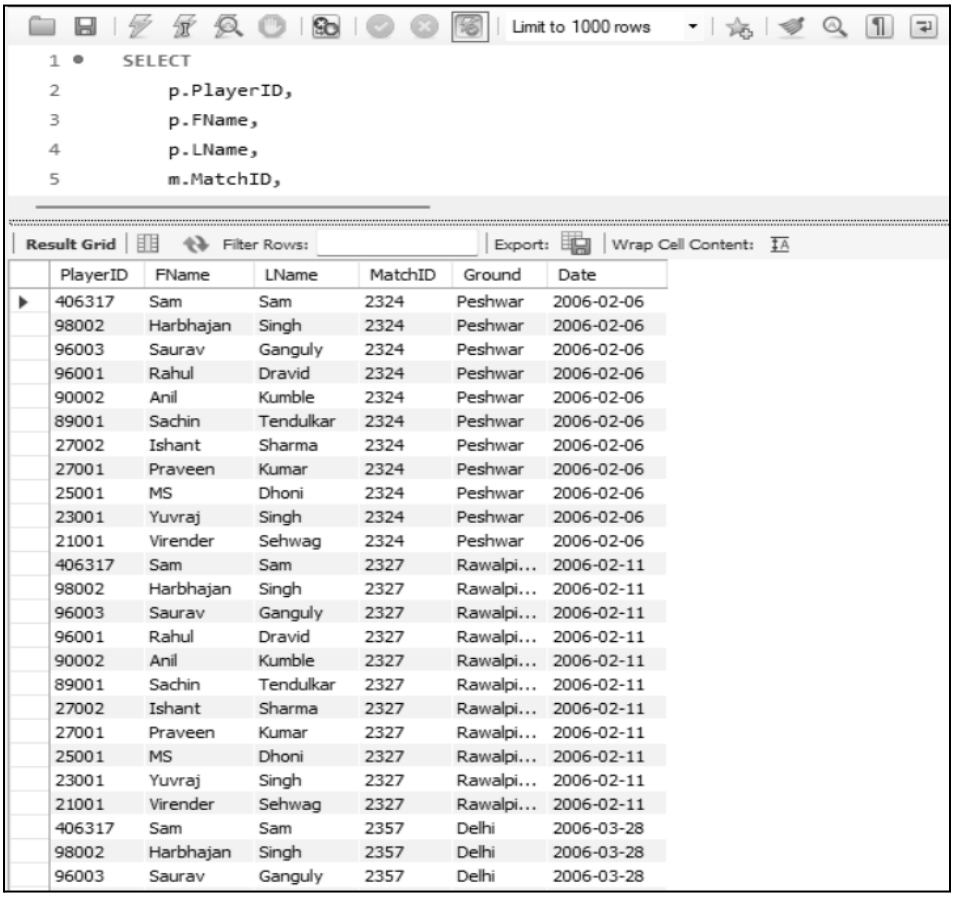
```
SELECT
m1.MatchID AS Match1,
m2.MatchID AS Match2,
m1.Ground,
m1.Date AS Date1,
m2.Date AS Date2
FROM odimatch m1
INNER JOIN odimatch m2
ON m1.Ground = m2.Ground
AND m1.MatchID < m2.MatchID
AND m1.Date <> m2.Date;
```



10. Cross Join:

Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

```
SELECT
p.PlayerID,
p.FName,
p.LName,
m.MatchID,
m.Ground,
m.Date
FROM player p
CROSS JOIN odimatch m
WHERE p.Country = 'India';
```



```
SELECT
    p.PlayerID,
    p.FName,
    p.LName,
    m.MatchID,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

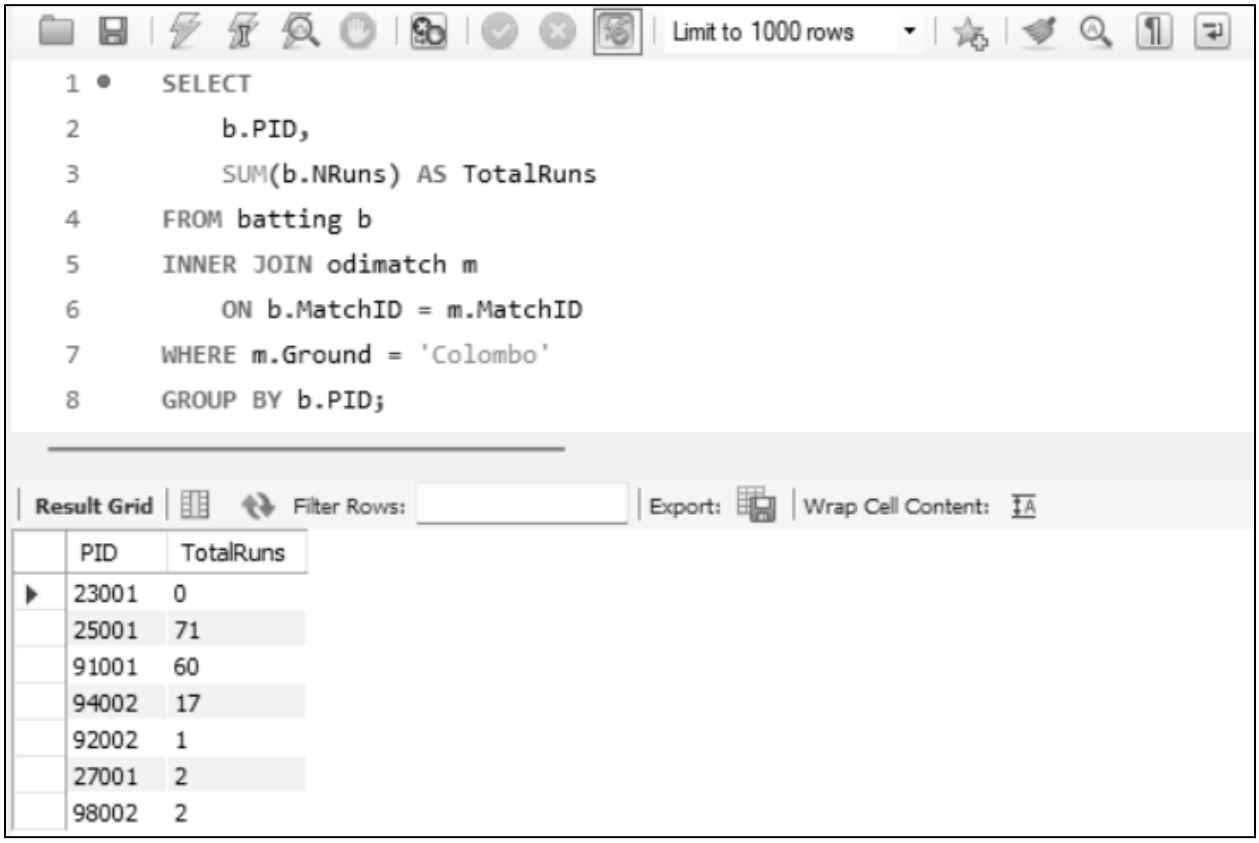
PlayerID	FName	LName	MatchID	Ground	Date
406317	Sam	Sam	2324	Peshwar	2006-02-06
98002	Harbhajan	Singh	2324	Peshwar	2006-02-06
96003	Saurav	Ganguly	2324	Peshwar	2006-02-06
96001	Rahul	Dravid	2324	Peshwar	2006-02-06
90002	Anil	Kumble	2324	Peshwar	2006-02-06
89001	Sachin	Tendulkar	2324	Peshwar	2006-02-06
27002	Ishant	Sharma	2324	Peshwar	2006-02-06
27001	Praveen	Kumar	2324	Peshwar	2006-02-06
25001	MS	Dhoni	2324	Peshwar	2006-02-06
23001	Yuvraj	Singh	2324	Peshwar	2006-02-06
21001	Virender	Sehwag	2324	Peshwar	2006-02-06
406317	Sam	Sam	2327	Rawalpi...	2006-02-11
98002	Harbhajan	Singh	2327	Rawalpi...	2006-02-11
96003	Saurav	Ganguly	2327	Rawalpi...	2006-02-11
96001	Rahul	Dravid	2327	Rawalpi...	2006-02-11
90002	Anil	Kumble	2327	Rawalpi...	2006-02-11
89001	Sachin	Tendulkar	2327	Rawalpi...	2006-02-11
27002	Ishant	Sharma	2327	Rawalpi...	2006-02-11
27001	Praveen	Kumar	2327	Rawalpi...	2006-02-11
25001	MS	Dhoni	2327	Rawalpi...	2006-02-11
23001	Yuvraj	Singh	2327	Rawalpi...	2006-02-11
21001	Virender	Sehwag	2327	Rawalpi...	2006-02-11
406317	Sam	Sam	2357	Delhi	2006-03-28
98002	Harbhajan	Singh	2357	Delhi	2006-03-28
96003	Saurav	Ganguly	2357	Delhi	2006-03-28

Many more total 200 rows affected

11. Natural Join vs Explicit:

Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

```
SELECT
b.PID,
SUM(b.NRuns) AS TotalRuns
FROM batting b
INNER JOIN odimatch m
ON b.MatchID = m.MatchID
WHERE m.Ground = 'Colombo'
GROUP BY b.PID;
```

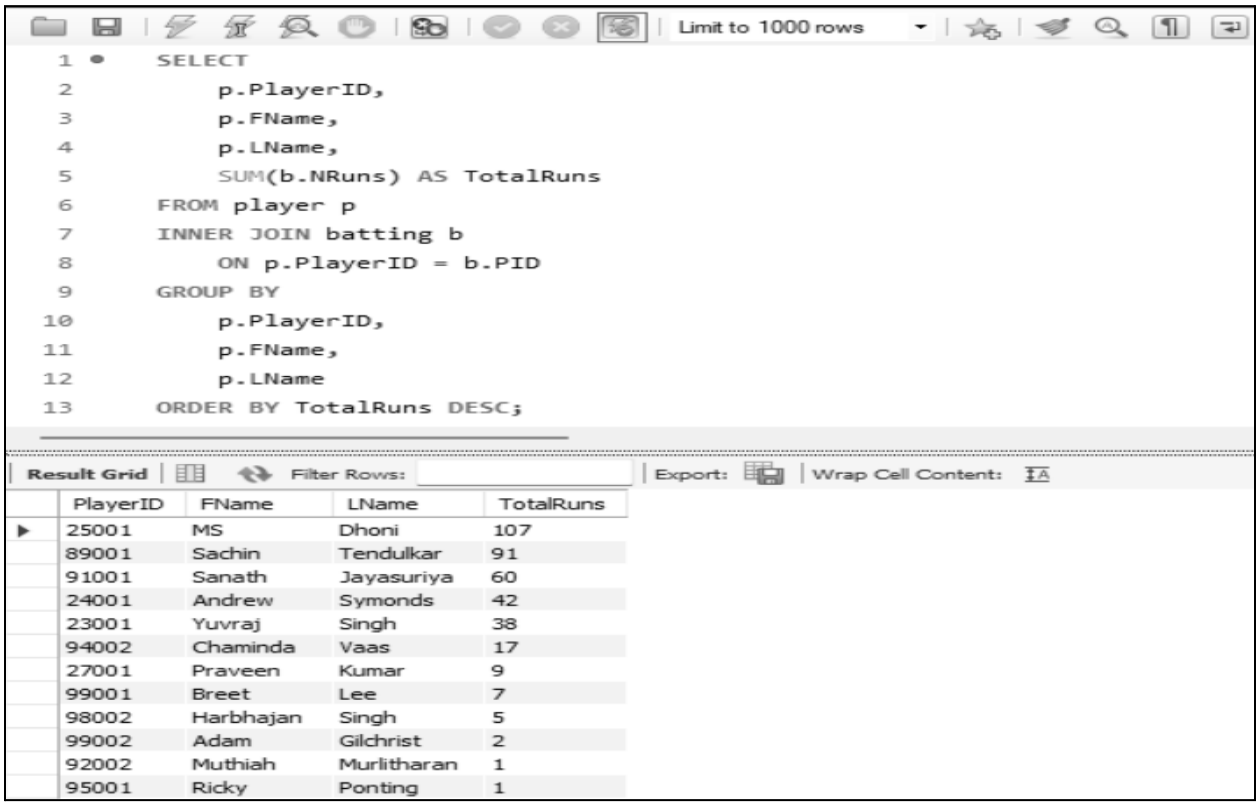


PID	TotalRuns
23001	0
25001	71
91001	60
94002	17
92002	1
27001	2
98002	2

12. Join with Aggregation:

Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

```
SELECT
p.PlayerID,
p.FName,
p.LName,
SUM(b.NRuns) AS TotalRuns
FROM player p
INNER JOIN batting b
ON p.PlayerID = b.PID
GROUP BY
p.PlayerID,
p.FName,
p.LName
ORDER BY TotalRuns DESC;
```

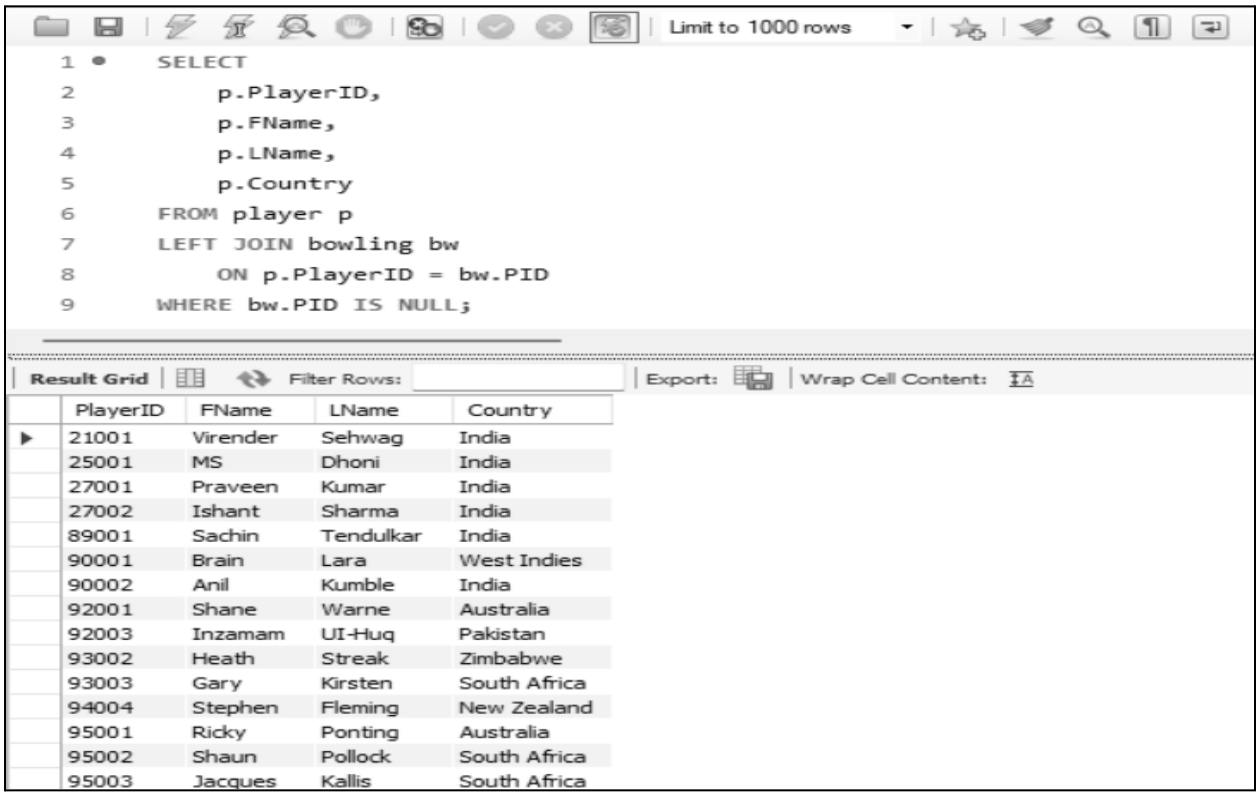


PlayerID	FName	LName	TotalRuns
25001	MS	Dhoni	107
89001	Sachin	Tendulkar	91
91001	Sanath	Jayasuriya	60
24001	Andrew	Symonds	42
23001	Yuvraj	Singh	38
94002	Chaminda	Vaas	17
27001	Praveen	Kumar	9
99001	Breet	Lee	7
98002	Harbhajan	Singh	5
99002	Adam	Gilchrist	2
92002	Muthiah	Murlitharan	1
95001	Ricky	Ponting	1

13. Left Join Anti-Join:

Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

```
SELECT
p.PlayerID,
p.FName,
p.LName,
p.Country
FROM player p
LEFT JOIN bowling bw
ON p.PlayerID = bw.PID
WHERE bw.PID IS NULL;
```



```
SELECT
    p.PlayerID,
    p.FName,
    p.LName,
    p.Country
FROM player p
LEFT JOIN bowling bw
    ON p.PlayerID = bw.PID
WHERE bw.PID IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Kumble	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa



96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa
96003	Saurav	Ganguly	India
97004	Daniel	Vettori	New Zealand
98001	Shahid	Afridi	Pakistan
99002	Adam	Gilchrist	Australia
99003	Michale	Vaughan	England
300311	Michale	Clark	NULL
406317	Sam	Sam	India